

COCI '18 Contest 3 #1 Magnus

Time limit: 1.0s **Memory limit:** 64M

Magnus lost a game of chess to Kile so he found comfort in competitive programming. Very soon, he heard of the iconic COCI competition and decided to try his luck there.

He wrote a mail to Kile: "*Dear Kile, please, prepare me for COCI. Magnus*".

Kile replied: "*You want to participate in COCI? All right, here's your warm-up task. A series of four consecutive letters of some word that make up the subword `HONI` (Croatian acronym for COCI) is called the HONI-block. I will send you the word of length N and you will throw out as many letters as you want (it might be none as well) so that in the end there are as many HONI-blocks as possible in the word. Kile*".

Magnus was very worried and asked you, COCI competitive scene, for help. Help him determine the maximum number of HONI-blocks he can get in the final word.

Input

The first line contains a word of length N ($1 \leq N \leq 100\,000$), consisting of uppercase letters of the English alphabet.

Output

In the first and only line, print out the required number of HONI-blocks.

Sample Input 1

```
MAGNUS
```

Sample Output 1

```
0
```

Sample Input 2

```
HHHHOOOONNNNIIII
```

Sample Output 2

1

Explanation for Sample Output 2

By throwing out three letters **H**, **O**, **N** and **I** Magnus can get the word **HONI**, which contains one HONI-block.

Sample Input 3

PROHODNIHODNIK

Sample Output 3

2